Exploratory Data Analysis Findings

Lead Time Distribution

The Lead Time distribution is positively skewed, indicating that most bookings are made relatively close to the arrival date, while a smaller number of bookings are made several months in advance.

Business Insight: Longer lead times may increase the likelihood of cancellations due to changing customer plans. Revenue managers should monitor early bookings and implement dynamic pricing strategies accordingly.

ADR Distribution

The ADR distribution contains several extreme values, indicating the presence of premium-priced bookings. Outlier treatment was applied to prevent these observations from disproportionately influencing the analysis.

Business Insight: Most bookings fall within a moderate price range, suggesting that standard room pricing drives the majority of hotel revenue.

Cancellation Rate Analysis

The visualization comparing cancellation status reveals a substantial proportion of canceled bookings.

Business Insight: Reducing cancellations can directly improve occupancy rates, revenue generation, and operational planning accuracy.

Market Segment Analysis

Different market segments exhibit varying booking and cancellation behaviors.

Business Insight: OTA customers tend to have different cancellation patterns compared to direct and corporate customers, highlighting the importance of segment-specific marketing and pricing strategies.